Chapter 11 — Special Theory of Relativity

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Q.1 — Multiple Choice Questions (with correct option and reason)

11.1. Relativistic mechanics gives results different from classical mechanics for objects moving with:

(a) low velocity
(b) velocity equal to sound
(c) velocity greater than sound
(d) velocity approaching that of light

Answer: (d) velocity approaching that of light — Relativistic effects become large near c.

11.2. If an observer moves in the same direction as a sound wave, the wave velocity seems:

(a) more
(b) less
(c) constant
(d) sum of the two velocities

Answer: (b) less — Relative speed is reduced.

11.3. If rest mass m_0 increases to m at high speed, kinetic energy is:

(a) $\frac{1}{2}mc^2$
(b) $\frac{1}{2}m_0c^2$
(c) $(m - m_0)c^2$
(d) $\frac{1}{2}(m - m_0)c^2$

Answer: (c) $(m - m_0)c^2$ — K = total energy − rest energy.

11.4. The speed of beam light of a moving car compared with its rest position is:

(a) greater
(b) less
(c) same
(d) zero

Answer: (c) same — Speed of light is invariant.

11.5. A photon moving at 0.9c has mass:

(a) 9.1×10^{-31} kg
(b) 1.67×10^{-27} kg
(c) 1.67×10^{-25} kg
(d) zero

Answer: (d) zero — A photon has zero rest mass and always travels at c.

The question's 0.9c is unphysical for a photon; its rest mass is zero.

Q.2 — Short Answer Questions

11.1. What is meant by inertial and non-inertial frames of reference?

An inertial frame is one in which Newton's first law holds; a body with no net force moves uniformly. A non-inertial frame is accelerating, so objects appear to accelerate without real forces.

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11.2. What are the two postulates of special relativity?

(1) The laws of physics are the same in all inertial frames. (2) The speed of light in free space is the same for all observers, regardless of source or observer motion.

11.3. Why is it impossible for a material particle to move with the speed of light?

Its relativistic mass increases without limit as v approaches c, so infinite energy would be required.

11.4. Does relativity contradict Newton's laws?

No. Newton's laws are the low-speed limit of relativity. When $v \ll c$, relativistic equations reduce to classical ones.

11.5. What is meant by proper time and proper length?

Proper time is the interval measured by a clock at rest relative to the events. Proper length is the length measured in the object's rest frame.

11.6. What is meant by relativistic mass, length and time?

Relativistic mass increases with speed, length contracts along motion, and time intervals dilate for a moving clock, all by the factor $\gamma = 1/\sqrt{(1 - v^2/c^2)}$.

11.7. Why does the mass of a moving object increase?

As speed increases, added energy contributes to inertia. The relativistic mass $m = m_0/\sqrt{(1 - v^2/c^2)}$ grows without limit near c.

11.8. All motion is relative. Is space-time absolute?

No. Space and time measurements depend on the observer, but the four-dimensional space-time interval is invariant.

11.9. Explain that the speed of light is an ultimate limit for any object.

As v approaches c, relativistic mass and required energy grow without limit, so no material object can reach or exceed c.

11.10. Give examples where special relativity has been verified.

Particle accelerators, lifetime dilation of fast muons, GPS clock corrections and mass–energy conversion in nuclear reactions all confirm relativity.

Constructed Response Questions

11.1 Speed of sound is affected by relative motion. Does this apply to light?

No. The speed of light in vacuum is the same for all inertial observers; only frequency and wavelength change via the Doppler effect.

11.2 Is it possible to see a star moving away at uniform velocity equal to c?

No material star can move at c. Also, light from a source receding at c would be infinitely red-shifted and could not reach us.

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11.3 If the speed of light were just 50 m s^{-1}, how would everyday events appear?

Relativistic effects would be obvious: moving objects would appear contracted, clocks would run slow, and mass would increase noticeably at ordinary speeds.

11.4 If the speed of light were infinite, what would relativity equations reduce to?

They would reduce to classical Newtonian equations, with no time dilation, length contraction or mass increase.

11.5 According to $E = mc^2$, is it possible to create a single electron from energy?

A single electron cannot be created alone because charge and lepton number must be conserved. Energy can create an electron–positron pair, with total charge zero.

Comprehensive / Long Questions

11.1 What is a frame of reference? Distinguish inertial and non-inertial frames with examples.

A frame is a coordinate system for measuring position and time. An inertial frame moves uniformly, e.g. a train at constant velocity; Newton's first law holds. A non-inertial frame accelerates, e.g. a braking car, where objects appear to lurch forward.

11.2 Describe Einstein's mass-energy equation. Why are its effects not observed in everyday life? Give consequences and examples.

$E = mc^2$ shows mass and energy are equivalent. Everyday mass changes are too tiny to notice because c^2 is enormous. Consequences include nuclear energy, pair production and annihilation, and stellar energy generation.

11.3 State Einstein's concept of space-time. Describe gravity according to this concept.

Space-time is a four-dimensional continuum combining space and time. Mass and energy curve space-time, and bodies follow geodesics in this curved geometry, which we observe as gravity.

Numerical Problems (fully worked)

11.1 An electron at 0.995c passes down a 500 m tube. How long does the tube appear to the electron?

Given: $L_0 = 500$ m, $v = 0.995c$

Formula: $L = L_0\sqrt{(1 - v^2/c^2)}$

$L = 500 \times \sqrt{(1 - 0.990025)} \approx 500 \times 0.1 = 50$ m.

Answer: 50 m

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11.2 A neutron disintegrates in 20 min on average. How long does it seem to exist at 0.8c?

Given: $t_0 = 20$ min, $v = 0.8c$

Formula: $t = t_0/\sqrt{(1 - v^2/c^2)}$

$t = 20/0.6 \approx 33.3$ min.

Answer: 33.3 min

11.3 A spaceship is 100 m at rest. What length is measured at 0.99c?

Given: $L_0 = 100$ m, $v = 0.99c$

Formula: $L = L_0\sqrt{(1 - v^2/c^2)}$

$L = 100 \times \sqrt{(1 - 0.9801)} \approx 100 \times 0.141 \approx 14$ m.

Answer: 14 m

11.4 Rest mass of electron 9.11×10^{-31} kg. Calculate rest-mass energy.

Given: $m_0 = 9.11\times10^{-31}$ kg

Formula: $E = m_0c^2$

$E = 9.11\times10^{-31} \times 9\times10^{16} \approx 8.2 \times 10^{-14}$ J ≈ 0.51 MeV.

Answer: 8.2×10^{-14} J or 0.51 MeV

11.5 An electron accelerated to 0.85c. Find total energy and kinetic energy in eV.

Given: $v = 0.85c$, rest energy = 0.511 MeV

Formula: $\gamma = 1/\sqrt{(1 - v^2/c^2)}$, $E = \gamma m_0c^2$, $K = (\gamma - 1)m_0c^2$

$\gamma \approx 1.906$.

$E \approx 0.97$ MeV.

$K \approx 0.46$ MeV.

Answer: 0.97 MeV, 0.459 MeV

11.6 At what speed is the mass of a proton tripled?

Given: $m = 3m_0$

Formula: $\gamma = 3$, $v = c\sqrt{(1 - 1/\gamma^2)}$

$v = c\sqrt{(1 - 1/9)} \approx 0.943c$.

Answer: 0.943c

11.7 A pendulum period is 3 s in its rest frame. What period is measured by an observer moving at 0.85c?

Given: $t_0 = 3$ s, $v = 0.85c$

Formula: $t = t_0/\sqrt{(1 - v^2/c^2)}$

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$\gamma \approx 1.906$, so $t \approx 5.72$ s.

Answer: ≈ 5.72 s

The book prints 9.68 s, which corresponds to about 0.95c rather than 0.85c.

11.8 A 0.5 kg ball is projected at 0.9c. What is its mass in flight?

Given: $m_0 = 0.5$ kg, $v = 0.9c$

Formula: $m = m_0/\sqrt{(1 - v^2/c^2)}$

$\gamma \approx 2.294$, $m \approx 1.15$ kg.

Answer: 1.15 kg

Textbook checks flagged in this chapter

- MCQ 11.5: a photon cannot move at 0.9c; its rest mass is zero.
- Numerical 11.7: book answer 9.68 s appears to use 0.95c; for 0.85c the result is about 5.72 s.

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